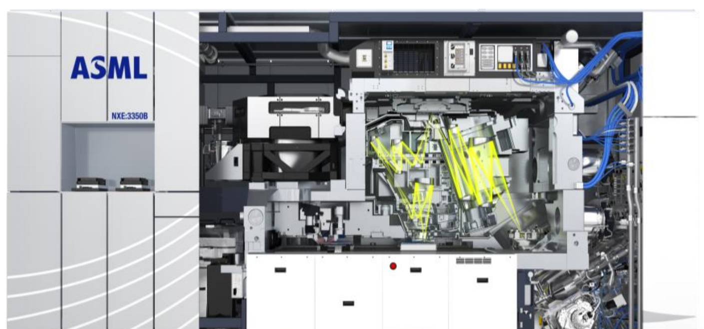
2MTEST : A semi-automatic tool for legacy test migration to MTEST

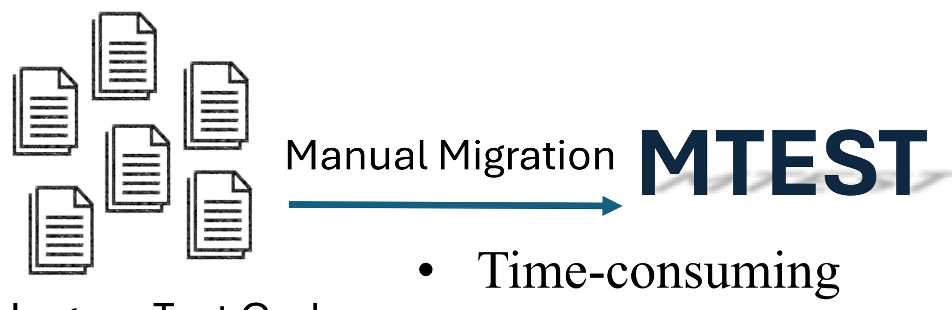
Introduction/ Context



Legacy Test Code:
- Initializing the whole twin scan software.
- Slow test run and high CPU usage

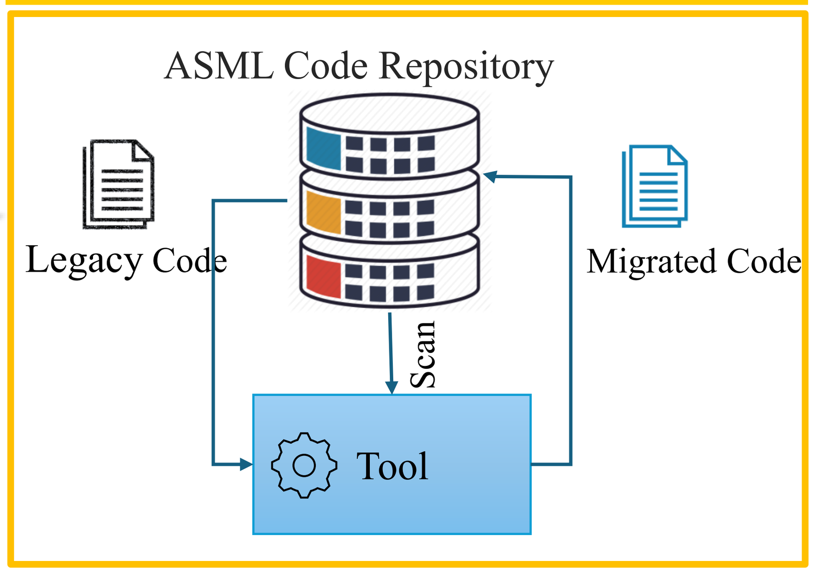
MTEST:
- ASML specific Module Testing framework.
- Facilitate testing in controlled and isolated environment
- Only necessary components are activated

Problem Statement

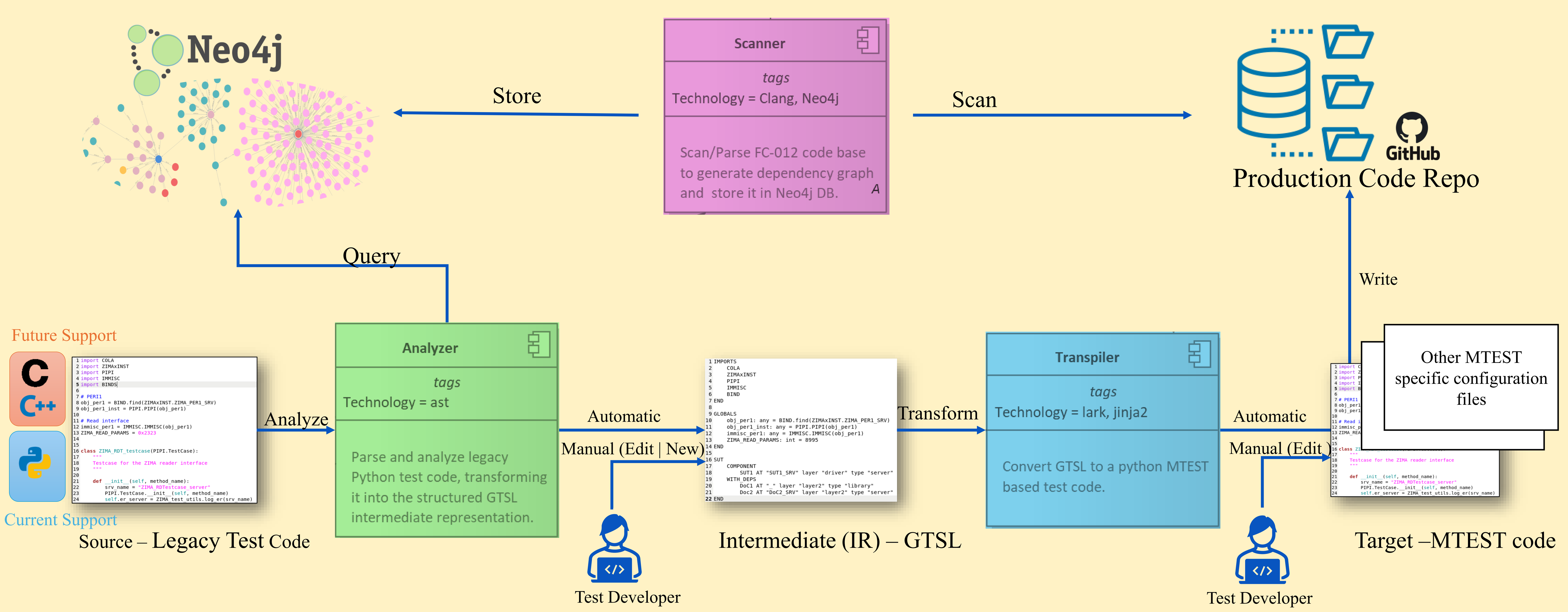


- Time-consuming
- Error-prone
- Costly

Solution Automation



Modular Migration Pipeline



Scanner: Knowledge Graph

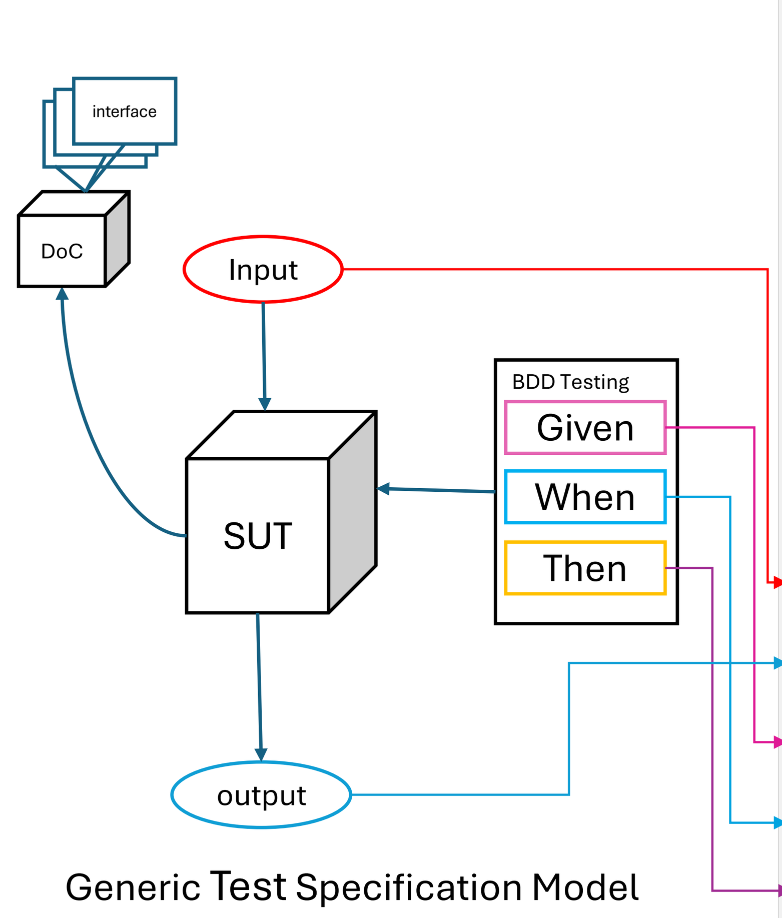
- **Pre-Processing**: Sanitizes *compile_commands .json*.
- **Hybrid Parsing**: Parsing C/C++ source code and ASML specific various configuration files using *libclnag* and regex.
- **Node/Hierarchical Discovery**: Creates a node for every entity (**Component**, **File**, **Function**, **Interface**) and build a global function definitions **symbol table**.
- **Dependency Linking**: Creates relationships (:**CALLS**, :**INCLUDES**, :**PART_OF**) using symbol table.
- **Output**: knowledge graph stored in Neo4j DB.
  - **Cypher** pseudo query => "Given *SUT function, what external components and interface are ultimately called*?"

Analyzer: Algorithm

1. Find all assertion calls
2. For each assertion extract actual and expected output
   - Track output variable
3. Backtrack from assertion to find assignments of the actual output from function call
4. Extract the input from each SUT calls
5. Segment the test case into Input/ Output Given/When/Then sections

Transpiler: GTSL – DSL

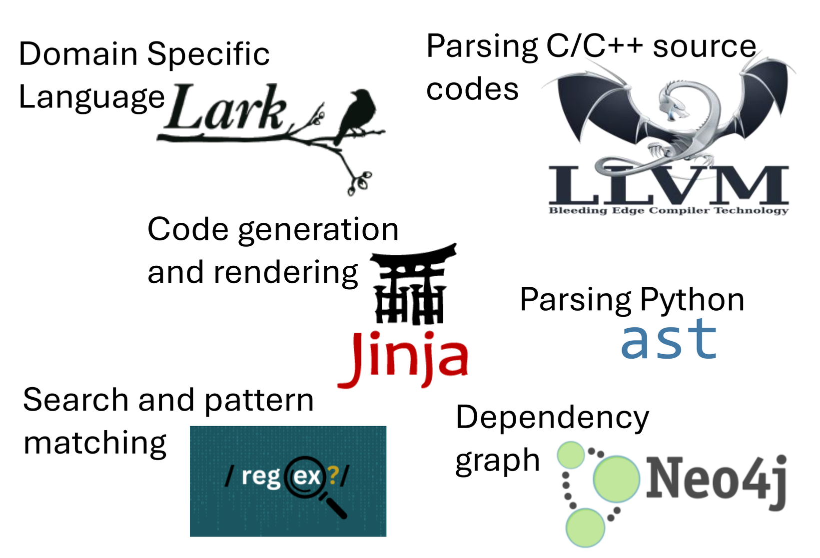
Generic Test Specification Language (**GTSL**)
- A domain-specific language (**DSL**) designed as an **intermediate representation** between legacy test code and the target MTEST based test.



Generic Test Specification Model

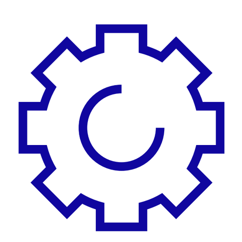
```
IMPORTS
    CCxIF
    IPPE
END
GLOBALS
    global_int: int = 123
END
SUT
    COMPONENT
        sut1 at "server_addr" layer "layer1" type "server"
    WITH_DEPS
        doc1 at "server_addr" layer "layer2" type "server"
        doc2 at "_" layer "layer3" type "library"
    END
END
DOUBLE
    INTERFACE MOKA FROM CMOT AT "SRV_ADDR1" IFTYPE "library"
    INTERFACE STUB1 FROM ZIMAxINST AT "STUB_ADDR" IFTYPE "mock"
END
SUITE TestSuiteExample
    SETUP
        call obj_per1_inst.terminate()
    END
    TEARDOWN
        set global_counter = 0
        comment "Clean up test environment"
    END
    HELPER
        FUNCTION check_status(obj: any) RETURNS bool
            return call obj.status()
        END
    CASE test_case_sum
        INPUT
            in reader_elem: any
            in scramble: int[] = [12,3,4]
        END
        OUTPUT
            out expected_out_1: float
        END
        GIVEN
            set temp = input1
            call obj_per1_inst.initialize(PIPI.INIT_LEVEL_1)
        END
        WHEN
            set result = call compute_sum(input1, input2)
        END
        THEN
            EXPECT_EQUAL result, 30
        END
    END
END
```

Technologies Used

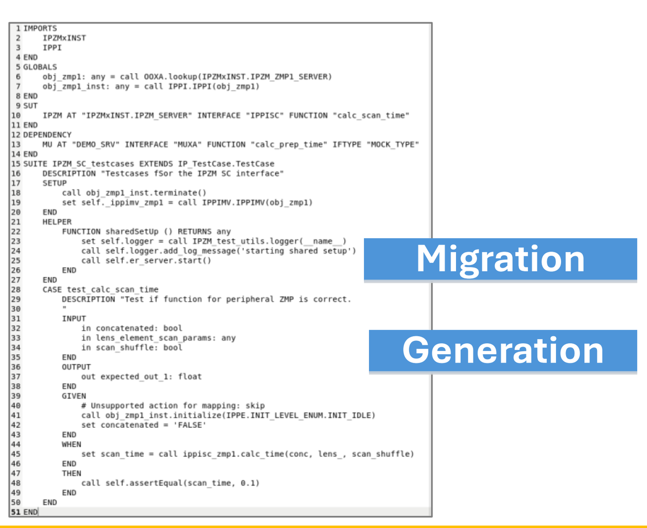


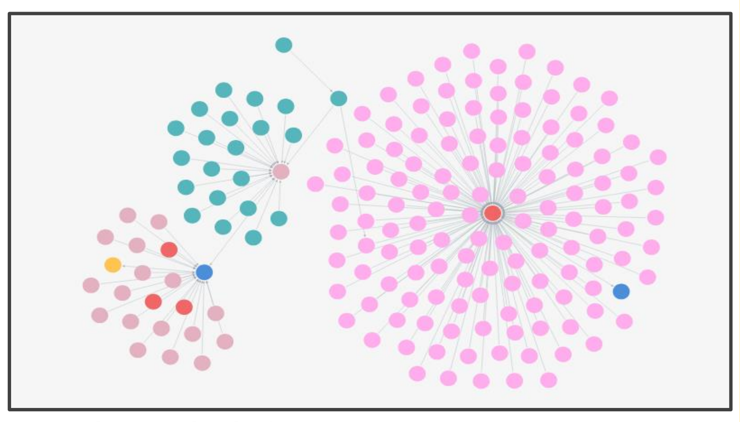
Value Added /Result

Prototype Tool	Domain Specific Language	Dependency Graph



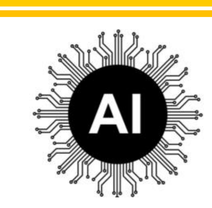
A tool that cuts 2 – 5 hours of manual migration into less 20 seconds





- Impact Analysis
- Circular Dependency Detection
- Visualization (Understanding)

Future Direction



Replacing the heuristic-based analyzer with AI/ML model.

Extending GTSL as test framework

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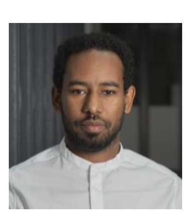

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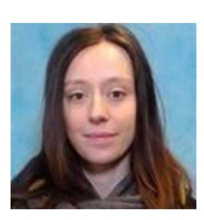

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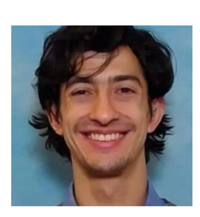

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